

$$\mathbf{p} = \hat{n} \tan(\omega/2)$$

$$R(\phi_1, \Phi, \phi_2)$$

$$\omega = 2\arccos(q_0) \quad \mathbf{q} = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k}$$

$$|\mathbf{q}| = 1$$

$$SO(3)$$

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$$S^3$$

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$$SO(3)$$

$$R_{ij} = \delta_{ij} \cos \omega + n_i n_j (1 - \cos \omega) - \varepsilon_{ijk} n_k \sin \omega$$

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$$R(\phi_1, \Phi, \phi_2)$$